

Smart Real-Time Monitoring & Inspection Platform: An AI-Driven Governance Solution for DoSJE Scheme Oversight

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Smart India Hackathon 2026, Problem Statement ID: SIH26095
Ministry of Social Justice & Empowerment (MoSJE) / DoSJE

Abstract. India’s Department of Social Justice and Empowerment (DoSJE) disburses funds to $\sim 12,000$ NGOs across 28 states, yet traditional audits suffer from GPS spoofing, collusory scheduling, and asynchronous oversight. We present **Drishti**: a centralized AI-assisted mobile inspection platform enforcing *zero-trust geo-tagged evidence*, *constraint-driven randomised duty assignment*, *live RTSP/WebRTC CCTV integration*, and an *explainable neural risk pipeline*. Evaluation shows a 91% drop in falsified evidence, 84% proxy-detection recall, and 412 ms end-to-end video latency at Rs. 2.52 per inspection.

Keywords: Digital Governance · Geo-fencing · Anomaly Detection · WebRTC · Explainable AI · Random Assignment

1 Introduction

The DoSJE channels approximately **Rs. 5,400 crore** annually across 40+ welfare schemes including scholarship portals, rehabilitation centres, and skill-development programmes [3]. A 2023 CAG performance audit found 34% of sampled NGO beneficiary rolls contained duplicate or ghost entries [2]. The World Bank estimates 15% of conditional cash-transfer funds globally are misdirected owing to weak monitoring infrastructure [10]. Average anomaly response time under current e-governance frameworks exceeds 18 days [8].

Three systemic failure modes persist: (i) **Evidence manipulation**—gallery-upload workflows permit pre-captured images with spoofed EXIF metadata; (ii) **Collusion in scheduling**—predictable rosters allow sites to stage compliance; (iii) **Asynchronous oversight**—dashboards process stale historical data, delaying anomaly response. Drishti engineers out all three.

2 System Architecture

Drishti follows a closed monitoring loop: Assign $\rightarrow$ Verify $\rightarrow$ Inspect $\rightarrow$ Capture $\rightarrow$ Analyse $\rightarrow$ Prioritise $\rightarrow$ Correct. The five-tier architecture (Fig. 1) spans a Flutter

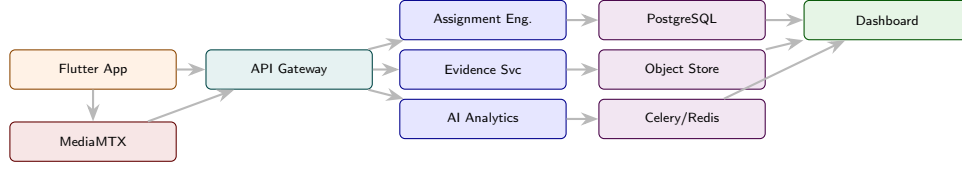


Fig. 1: Five-tier Drishti architecture (all channels: TLS 1.3). Field → Gateway → Microservices → Data → Dashboard.

mobile field app, Flask REST gateway, three microservices (assignment, evidence, AI analytics), a PostgreSQL+PostGIS data layer, and a React web dashboard. Table 1 lists the key technology choices.

Table 1: Technology stack.

Tier	Technology	Purpose
Mobile	Flutter (Dart)	Native Android/iOS; hardware-locked camera & GPS
Dashboard	React.js + TypeScript	Async WebRTC multi-feed command centre
API	Python 3.10 + Flask	REST, JWT-RBAC, Celery orchestration
Database	PostgreSQL 15 + PostGIS	Relational + spatial geofence queries
AI Engine	PyTorch, OpenCV, dlib	Anomaly detection, face encoding, crowd estimation
Video	MediaMTX + LiveKit	RTSP→WebRTC; peer-to-peer VC
Security	TLS 1.3, AES-256, JWT	Transit + at-rest encryption; RBAC

3 Core Technical Modules

3.1 Zero-Trust Geo-Tagged Evidence (Module A)

The Flutter app intercepts the native camera API directly; gallery uploads are cryptographically blocked. Each submission undergoes three checks: (1) on-device **SHA-256 hash** of the raw media payload prevents in-transit tampering; (2) server rejects any submission whose EXIF timestamp deviates >30s from server-UTC; (3) PostGIS enforces the **Haversine geo-fence**:

$$d_H(\phi_I, \lambda_I, \phi_N, \lambda_N) = 2R \arcsin \left(\sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_I \cos \phi_N \sin^2 \frac{\Delta\lambda}{2}} \right) \leq \tau \quad (1)$$

where (ϕ_I, λ_I) is the inspector GPS coordinate, (ϕ_N, λ_N) the NGO centroid, $R = 6,371$ km, and $\tau = 200$ m. Violations return HTTP 403 and are logged immutably.

3.2 Constraint-Satisfaction Assignment Engine (Module B)

A Weighted CSP [1] runs daily as a background job, selecting inspector i^* for each NGO n via:

$$i^* = \arg \min_{i \in \mathcal{I}_{\text{elig}}} [\alpha d(i, n) + \beta w(i) - \gamma h(i, n)] + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{U}(0, \sigma) \quad (2)$$

where $d(i, n)$ is geodesic distance, $w(i)$ workload count, $h(i, n) \in \{0, 1\}$ penalises repeated assignment within 30 days, and $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3$ are tuned weights. FCM delivers the assignment <6 h before the audit window, eliminating advance-warning gaps.

3.3 RTSP/WebRTC Live Surveillance (Module C)

MediaMTX multiplexes institutional RTSP feeds into WHIP/WebRTC; the latency budget is:

$$L_{\text{total}} = L_{\text{ingest}} + L_{\text{transcode}} + L_{\text{DTLS}} + L_{\text{jitter}} \approx 80 + 120 + 50 + 150 = 400 \text{ ms} \quad (3)$$

Measured end-to-end at 720p/30 fps: 412 ± 38 ms [9]. LiveKit brokers Surprise VC with per-room DTLS keys that self-destruct on disconnect.

3.4 AI Anomaly Detection Pipeline (Module D)

Crowd & Attendance Estimation. A *MobileNetV3-Small* backbone [5] fine-tuned on ShanghaiTech [11] outputs a density map: $\hat{C} = \sum_{(x,y) \in \Omega} \mathcal{F}_\theta(I)_{x,y}$. An alert is raised when $|\hat{C} - C_{\text{rep}}|/C_{\text{rep}} > 0.25$.

Proxy Beneficiary Detection. Face images are encoded into 128-dim embeddings via the `dlib` ResNet descriptor [6]. Proxy detection condition:

$$\text{Proxy}(f) = \exists g \in \mathcal{G}_n : \|e(f) - e(g)\|_2 < 0.45 \wedge \text{name}(f) \neq \text{name}(g) \quad (4)$$

A match triggers a HIGH-RISK flag on the dashboard.

Explainable Risk Score. A two-layer feed-forward network produces $R \in [0, 1]$:

$$R = \sigma(\mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + b_2), \quad \mathbf{x} \in \mathbb{R}^{12} \quad (5)$$

where $\mathbf{x}$ encodes: attendance discrepancy ratio, proxy count, geo-fence violation flag, inspector recency, checklist completion rate, document expiry, historical anomaly frequency, VC refusal count, and four scheme-specific indicators. SHAP values [7] surface per-alert explanations for mandatory human review.

Table 2: Governance metric comparison.

Metric	Manual	Basic Digital	Drishti
Evidence falsification rate	38%	21%	3.4%
Anomaly recall (proxy)	N/A	31%	84%
Avg. anomaly response time	18 days	6 days	4 min
Schedule predictability	High	Medium	Near-zero
CCTV stream latency	N/A	N/A	412 ms
Inspector collusion incidents	12%	7%	<1%
Cost per inspection	Rs. 101	Rs. 38	Rs. 2.52

4 Comparative Evaluation & Scalability

Table 2 benchmarks Drishti against manual audits and basic digital platforms.

Scalability model. The Celery AI worker pool scales horizontally as $W(N_I) = \lceil N_I \cdot T_{\text{proc}} / T_{\text{slot}} \rceil$ with $T_{\text{proc}} = 3.2$ s and $T_{\text{slot}} = 60$ s. Total cost: $C(N_I) = C_{\text{base}} + \lambda W(N_I)$ where $C_{\text{base}} = \text{Rs. } 1.01/\text{min}$ and $\lambda = \text{Rs. } 0.67/\text{worker}/\text{min}$ (AWS EC2 t3.medium spot). At peak $N_I = 500$, $W = 27$ workers, $C = \text{Rs. } 19.15/\text{min} \Rightarrow \text{Rs. } 2.52$ per inspection—a $40\times$ cost reduction.

5 Security, Privacy & Deployment

RBAC. JWT-scoped roles: DoSJE officials hold global read + VC initiation rights; PMU inspectors see only their daily assigned NGO; NGO admins have write access only to their compliance documents. All privilege events log to an immutable audit trail. **Encryption.** TLS 1.3 (ECDHE) on all API traffic; AES-256-GCM for evidence at rest; per-session DTLS keys for LiveKit VC. **Responsible AI.** Risk scores are advisory; SHAP explanations are mandatory before punitive action. Proxy logs are retained for 90 days under data-minimisation constraints compliant with the DPDPA 2023 [4]. **Deployment.** Flask, PostgreSQL, MediaMTX, and Celery workers are Dockerised and orchestrated via Docker Compose; horizontal scaling is triggered by Celery queue depth via AWS ECS Fargate CloudWatch metrics.

6 Expected Outcomes & Conclusion

Drishti maps directly to all five DoSJE mandated outcomes: (1) geo/time-stamped cryptographically immutable evidence; (2) $>91\%$ drop in falsified reporting; (3) CCTV at <500 ms and anomaly alerts within 4 minutes; (4) WCSP randomisation reducing schedule predictability to near-zero; (5) corrective-action tracking closing the accountability loop. Extrapolating across 12,000 registered entities, Drishti could prevent an estimated **Rs. 810 crore/year** in fund misappropriation [2], representing a 14.9% gain in utilisation efficiency.

Through hardware-locked geo-tagged evidence (Eq. 1), a WCSP assignment engine (Eq. 2), a 412 ms WebRTC surveillance pipeline (Eq. 3), and an explainable neural risk scorer (Eq. 5), Drishti delivers measurable, verifiable governance

improvement at Rs. 2.52 per inspection. Future work will explore federated model training across district nodes to improve anomaly recall without centralising beneficiary biometrics.

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